

PHASE 1 - SYSTEM SPECIFICATION (SSOT)

CFO Bot: Cloud Cost Estimation System

1. System Overview

The CFO Bot is a deterministic cloud cost calculator that estimates the monthly and long-term cost of running a chatbot application.

The system is based on:

- usage inputs
- abstract AI model tiers
- infrastructure pricing

2. System Scope

The system SHALL estimate costs for:

- AI inference (token-based usage)
- Hosting infrastructure
- Storage
- Bandwidth
- Database operations

3. AI Model Tiers (Simplified & Defendable)

Instead of real-world models, the system uses **abstract tiers** representing computational complexity.

3.1 Model Definitions

Model	Price per 1M tokens
Lite	\$0.10
Standard	\$1.00
Pro	\$3.00

3.2 Effective Rate

$effective_rate = base_rate$

4. Infrastructure Components

4.1 Hosting (Fixed Monthly Cost H)

Option	Cost
Free Tier	\$0
Basic Server	\$5
Standard Server	\$10

4.2 Storage

$storage_cost = storage_gb \times 0.02$

4.3 Bandwidth

$bandwidth_cost = bandwidth_gb \times 0.05$

4.4 Database

$db_cost = (db_reads \times 0.0001) + (db_writes \times 0.0002)$

5. Input Parameters

Parameter	Description
users	active users
messages_per_user	messages per day
tokens_per_message	tokens per request
model	selected AI tier
hosting_type	selected hosting

storage_gb	storage usage
bandwidth_gb	network usage
db_reads	daily reads
db_writes	daily writes
months	projection period
growth_rate	monthly growth

6. Core Mathematical Model

6.1 Monthly Requests

$$\text{Req} = \text{users} \times \text{messages_per_user} \times 30$$

6.2 Token Usage

$$\text{Tokens} = \text{Req} \times \text{tokens_per_message}$$

6.3 AI Cost

$$\text{AI_cost} = (\text{Tokens} / 1,000,000) \times \text{effective_rate}$$

6.4 Database Monthly Usage

$$\text{monthly_reads} = \text{db_reads} \times 30$$

$$\text{monthly_writes} = \text{db_writes} \times 30$$

6.5 Infrastructure Cost

$$\text{Infra_cost} = H + \text{storage_cost} + \text{bandwidth_cost} + \text{db_cost}$$

6.6 Total Monthly Cost

$$\text{C_mo} = \text{AI_cost} + \text{Infra_cost}$$

7. Long-Term Projection

$$\text{Total_n} = \sum_{(i=1 \rightarrow n)} [\text{C_mo} \times (1 + g)^{(i-1)}]$$

Where:

- n = number of months
- g = growth rate

8. Constraints & Validation

- $\text{users} \geq 1$
- all inputs ≥ 0
- $\text{growth_rate} \in [0,1]$
- $\text{months} \leq 60$
- model must be selected

9. Edge Case Handling

Case	Behavior
$\text{users} = 0$	error
$\text{tokens} = 0$	$\text{AI_cost} = 0$
$\text{storage} = 0$	$\text{storage_cost} = 0$
$\text{bandwidth} = 0$	$\text{bandwidth_cost} = 0$
invalid input	validation error

10. Architecture

10.1 Frontend

- HTML
- CSS
- JS

10.2 Pricing Engine

Embedded JavaScript module in index.html

Responsibilities:

- all formulas
- no UI logic
- reusable & testable

10.3 Deployment

- Firebase Hosting (required)

11. UI/UX Requirements

Layout

- Top → model & hosting selection
- Middle → input fields / sliders
- Bottom → **TOTAL COST (highlighted)**

Design

- minimalistic
- clean typography
- smooth transitions

Output

Display:

- AI cost
- Infrastructure cost
- Total cost

12. Testability Requirements

Example Test Cases

Scenario	Expected
users=100	valid cost
model=Lite	lower AI cost
storage=0	0
invalid input	error

13. Non-Functional Requirements

- response < 1s
- deterministic output
- no external APIs

high readability

14. Assumptions

- 30 days per month
- all users active daily
- linear scaling